



2026 **YEAR12**
TRIAL
HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

**General
Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black pen.
- Calculators approved by NESA may be used.
- Answer all questions in the booklet provided.
- Marks may be deducted for careless or badly arranged work
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.
- A reference sheet is provided at the back of this paper

**Total marks:
100**

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II – 90 marks (pages 6 – 13)

- Attempt Questions 11 – 16
- Allow about 2 hour 45 minutes for this section.

Section I

10 marks

Attempt questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer page in your booklet to answer Questions 1 – 10.

- 1) Consider the following statement:

“If n is a perfect square, then n is not 2 more than a multiple of 3.”

Which of the following is the negation of this statement?

- A) If n is not 2 more than a multiple of 3, then n is a perfect square.
- B) n is a perfect square and n is 2 more than a multiple of 3.
- C) If n is 2 more than a multiple of 3, then n is not a perfect square.
- D) n is not a perfect square and n is 2 more than a multiple of 3.

- 2) Which of the following expressions is equivalent to $\int \frac{x^3 dx}{x^2 + x + 1}$?

- A) $\frac{x^2}{2} - x + \tan^{-1} \frac{2x+1}{\sqrt{3}} + c$
- B) $\frac{x^2}{2} - x + \tan^{-1} \frac{4x+2}{3} + c$
- C) $\frac{x^2}{2} - x + \frac{4}{3} \tan^{-1} \frac{4x+2}{3} + c$
- D) $\frac{x^2}{2} - x + \frac{2}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} + c$

- 3) A polynomial $P(z)$ has real coefficients. It is known that:

$$P(1+2i) = 0$$

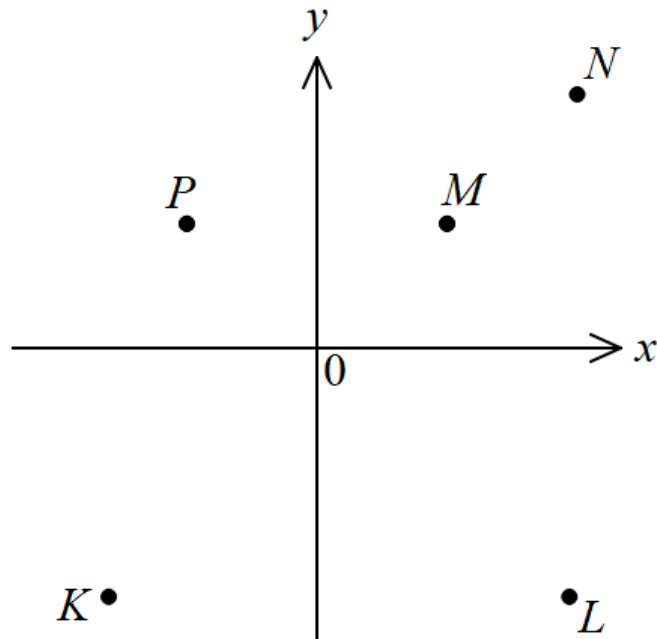
$$P'(1+2i) = 0$$

$$P(-3) = 0$$

What is the minimum possible degree of $P(z)$?

- A) 3
- B) 4
- C) 5
- D) 6

- 4) In the Argand diagram the point P represents a complex number. When this number is multiplied by $\frac{(1+i)^4}{(1-i)^2}$ it gave a new complex number.



Which of the following points represents the new complex number?

- A) K
 - B) L
 - C) M
 - D) N
- 5) Consider three vectors $\underline{a} = 2\underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{b} = -3\underline{i} + 4\underline{j} - \underline{k}$ and $\underline{c} = 13\underline{i} + 10\underline{j} + \underline{k}$. Which of the following statements is true?
- A) $\underline{b}$ and $\underline{c}$ are perpendicular to $\underline{a}$
 - B) $\underline{b}$ and $\underline{a}$ are perpendicular to $\underline{c}$
 - C) $\underline{c}$ is only perpendicular to $\underline{a}$
 - D) $\underline{c}$ is only perpendicular to $\underline{b}$

- 6) Consider the position vector of a particle $\underline{r} = t^2 \underline{i} + 2 \cos t \underline{j} + 2 \sin t \underline{k}$. Which of the following statements best describes the motion of the particle?
- A) A spiral about the z axis in an anticlockwise direction.
- B) A spiral about the x axis in an anticlockwise direction.
- C) A spiral about the z axis in a clockwise direction.
- D) A spiral about the x axis in a clockwise direction.
- 7) The points A and B have coordinates $(4, 3, -5)$ and $(2, -7, 6)$ respectively. Point P lies on the interval AB such that $PA : PB = 1 : 2$. What are the coordinates of P ?
- A) $\left(\frac{10}{3}, -\frac{1}{3}, -\frac{4}{3}\right)$
- B) $\left(\frac{14}{3}, \frac{19}{3}, -\frac{26}{3}\right)$
- C) $\left(\frac{4}{3}, -\frac{31}{3}, -\frac{29}{3}\right)$
- D) $\left(\frac{8}{3}, -\frac{11}{3}, \frac{7}{3}\right)$
- 8) Which of the following is a necessary and sufficient condition for function $f(x)$ to satisfy for all positive values of a ?
- $$\int_0^a [f(x-a) + f(a-x)] dx = 0.$$
- A) $f(x)$ is increasing.
- B) $f(x)$ is decreasing.
- C) $f(x)$ is even.
- D) $f(x)$ is odd.

- 9) A particle starts moving along a straight line from rest at the origin. At time t , its velocity is v and its displacement from a fixed origin is x .

If $\frac{dv}{dx} = \frac{2}{3v}$, which of the following best describes the particle's acceleration and velocity?

- A) Constant acceleration and constant velocity.
- B) Constant acceleration and decreasing velocity.
- C) Constant acceleration and increasing velocity.
- D) Increasing acceleration and increasing velocity.

- 10) A complex number ω satisfies $\omega^2 + \frac{1}{1+\omega^2} + \omega = 0$. Which of the following is the correct statement about ω^{2026} ?

- A) $\omega^{2026} = \omega$
- B) $\omega^{2026} = \omega^2$
- C) $\omega^{2026} = \omega^3$
- D) $\omega^{2026} = \omega^4$

End of Section I

Section II (90 marks)**Attempt Questions 11 - 16****Allow about 2 hour 45 minutes for this section**

Answer each question on the appropriate pages of your answer booklet. Extra paper is available.
Your responses should include relevant calculations and/or mathematical reasoning.

Question 11 (15 marks)

a) Given that z and w are two complex numbers, where $z = 1 + ai$ and $w = a + i$, $a \in \mathbb{R}$. Find:

i) $z + \bar{w}$. 1

ii) $\text{Arg}(zw)$. 2

iii) For what value(s) of a , will $\frac{z}{w}$ be purely real? 2

b) If $\underline{a} + \underline{b} = \underline{i}$ and $\underline{b} = 3\underline{i} + 2\underline{j} - 2\underline{k}$, what is the value of $|\underline{a}|$? 2

c) Find $\int \frac{x^3}{x^2 + 9} dx$. 2

d) Consider the statement P .

P : For all integers n , if $n^2 - 6n + 5$ is even, then n is odd.

i) Write down the contraposition of P . 1

ii) Hence prove that the proposition P is true. 2

e) Solve $z^2 - z + (4 - 2i) = 0$, giving your answers in the form of $x + yi$ where x and y are real numbers. 3

Question 12 (15 marks)

- a) The polynomial $P(z) = z^4 - 8z^3 + pz^2 + qz - 80$ has a root $3 + i$, where p and q are real numbers.

i) Find all the roots of $P(z)$.

3

ii) Write $P(z)$ as a product of two real quadratic factors.

1

- b) Show that $\frac{a^2 + b^2}{a - b} \geq 2\sqrt{2}$ for $a > b$ and $ab > 1$.

2

- c) The lines l_1 and l_2 are given by the equations:

3

$$l_1 : \vec{r} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + s \begin{pmatrix} 2 \\ \ln a \\ \ln a \end{pmatrix}, \quad l_2 : \vec{r} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + t \begin{pmatrix} -3 \\ \ln a \\ 1 \end{pmatrix}, \text{ where } s \in \mathbb{R}, t \in \mathbb{R} \text{ and } a \text{ is a positive}$$

constant.

Given that l_1 and l_2 are perpendicular, what are the values of a ?

- d) By using the substitution $t = \tan \theta$, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \frac{4}{5 - 3 \cos 2\theta} d\theta$.

3

- e) A particle moves along the x axis in a simple harmonic motion. Its displacement x cm from the origin at time t seconds is given by $x = 5 - 4 \cos^2 2\pi t$.

i) Find the centre and the amplitude of the motion.

2

ii) Find the distance travelled by the particle in the first 5 seconds.

1

Question 13 (14 marks)

a) Prove by contradiction that $P(x) = 120 + 120x + 60x^2 + 20x^3 + 5x^4 + x^5$ has no double roots. **2**

b) Let $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$, for $n \geq 0$.

i) Show that $I_{n+1} = \frac{1}{2n} \left[\frac{1}{2^n} + (2n-1)I_n \right]$, for $n \geq 1$. **3**

ii) Hence, evaluate I_3 . **2**

c) A particle of mass 1kg is projected from the origin with initial speed $V \text{ ms}^{-1}$ at an angle α to the horizontal plane.

The forces acting on the particle are weight force and air resistance of $5\mathbf{v}$, where $\mathbf{v}$ is the velocity vector of the particle at time t seconds after the particle is projected.

Let the acceleration due to gravity be $g = 10 \text{ ms}^{-2}$.

i) Show that $\mathbf{v}(t) = \begin{pmatrix} Ve^{-5t} \cos \alpha \\ (V \sin \alpha + 2)e^{-5t} - 2 \end{pmatrix}$. **3**

ii) Given that $\mathbf{v}(1) = \begin{pmatrix} 250e^{-5} \\ (250\sqrt{3} + 2)e^{-5} - 2 \end{pmatrix}$, find V and α . **2**

iii) Show that the ratio of the horizontal velocity at the origin to the horizontal velocity at the maximum height is $(1 + 125\sqrt{3}) : 1$. **2**

Question 14 (17 marks)

a) Let a , b and c be three positive real numbers. It is known that $a + b + c = 1$.

i) Prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$. 2

ii) Prove that $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \geq 8$. 3

b) A particle P of mass m kg is released from rest and falls under its own weight. The particle is subject to air resistance of magnitude mkv^2 newtons, where $v \text{ ms}^{-1}$ is the velocity of the particle after it has travelled x metres and k is a positive constant.

i) Show that $v^2 = (1 - e^{-2kx}) \frac{g}{k}$. 3

ii) Determine the terminal velocity V_T of the particle. 1

iii) Show that the particle attains a speed of $\frac{1}{2}V_T$ after it has fallen a distance of $\frac{1}{2k} \ln\left(\frac{4}{3}\right)$ metres. 2

c) i) Show that $\frac{1}{x\sqrt{x^{2n}-1}} = \frac{x^{-n-1}}{\sqrt{1-x^{-2n}}}$. 1

ii) Hence, or otherwise, find $\int_2^4 \frac{dx}{x\sqrt{x^{2n}-1}}$, $n = 1, 2, 3, \dots$. 3

iii) Show that $\int_2^4 \frac{dx}{x\sqrt{x^{2n}-1}} > \frac{1}{n} \left(\frac{2^n - 1}{4^n} \right)$. 2

Question 15 (14 marks)

- a) The acceleration of a particle is $\ddot{x} = 4x^3 - 16x$, where x is the displacement of the particle from a fixed point O after t seconds, in metres.

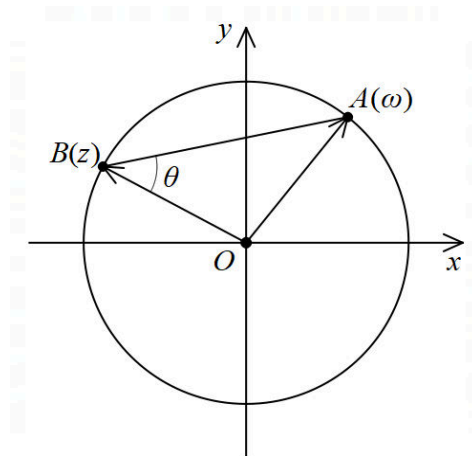
Initially the particle is at O and has a velocity of $v_0 \text{ ms}^{-1}$.

- i) Show that the quantity $(\dot{x})^2 - 2x^4 + 16x^2$ does not change with time. 2

- ii) Show that when $v_0 = \sqrt{\frac{31}{8}}$, the particle remains in the region $-\frac{1}{2} \leq x \leq \frac{1}{2}$. 2

- iii) State, with reason, whether the particle is moving in simple harmonic motion. 1

- b) The diagram shows two distinct points A and B that represent the complex numbers w and z respectively. These points lie on the circle with centre O and radius r . Let $\angle OBA = \theta$.

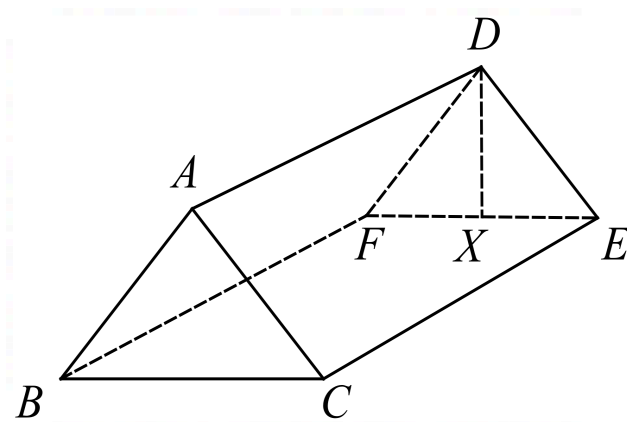


- i) Show that $z = w(-\cos 2\theta + i \sin 2\theta)$. 2

- ii) Write down the complex number represented by $\overline{AB}$ and deduce that $|z + w| = \tan \theta |z - w|$. 2

Question 15 continues on the next page

- c) The diagram shows the triangular prism $ABCDEF$, where $BCEF$ is a rectangle. The triangles ABC and DEF are equilateral.



Let $|\overrightarrow{AB}| = m$ and $|\overrightarrow{CE}| = n$. X is the midpoint of $\overrightarrow{EF}$.

- i) Use vector methods to show that $\overrightarrow{XD}$ is perpendicular to $\overrightarrow{EF}$.

3

- ii) Show that $|\overrightarrow{BX}| + |\overrightarrow{BF}|$ is less than the perimeter of the rectangle $BCEF$.

2

End of Question 15

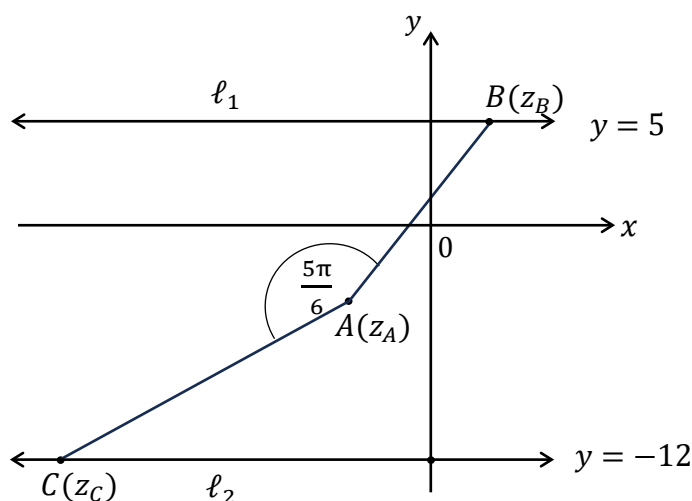
Question 16 (15 marks)

a) Use a suitable substitution to evaluate $\int_0^{\frac{\pi}{4}} \frac{1}{\cos x \sqrt{\sec x - \tan x}} dx$. 4

- b) The point A , which represents the complex number $z_A = -5 - 4i$, lies between two parallel lines ℓ_1 and ℓ_2 . 3

The equations of ℓ_1 and ℓ_2 are $y = 5$ and $y = -12$ respectively.

The point B , which represents the complex number z_B , lies on ℓ_1 . The point C , which represents the complex number z_C , lies on ℓ_2 such that $AC = 2AB$ and $\angle BAC = \frac{5\pi}{6}$.



Find the exact value of the complex number z_C .

Question 16 continues on the next page

c) Let $z_0, z_1, z_2, \dots, z_n$ be complex numbers such that

$$(k+1)z_{k+1} - i(n-k)z_k = 0 \text{ for all } k \in \{0, 1, 2, \dots, n-1\}.$$

i) Prove, by mathematical induction that $z_r = i^r \binom{n}{r} z_0$ for all integers $r \geq 0$.

3

It is known that the complex numbers form the identity $z_0 + z_1 + z_2 + \dots + z_n = 2^n$.

ii) Hence, show that $z_0 = (1-i)^n$.

2

It is known that $\sum_{m=0}^n \binom{n}{m}^2 = \binom{2n}{n}$. (Do NOT prove this)

iii) Hence or otherwise show that $|z_0|^2 + |z_1|^2 + |z_2|^2 + \dots + |z_n|^2 \leq \frac{(3n+1)^n}{n!}$.

3

End of Exam

X2 TRIAL 2026 Solutions

Section I - Multiple Choice Questions (1 - 10)

1) B

6) B

2) D

7) A

3) C

8) D

4) D

9) C

5) B

10) A

Suggestion Solutions

1) $\neg(P \Rightarrow Q)$ is $P \wedge \neg Q$

P: n is a perfect square

Q: n is not 2 more than a multiple of 3.

$P \wedge \neg Q$: n is a perfect square and n is 2 more than a multiple of 3.

$\therefore$ B is correct answer.

2) $\int \frac{x^3 dx}{x^2 + x + 1}$

$$\therefore \int \left((x-1) + \frac{1}{x^2+x+1} \right) dx$$

$$\int x dx - \int dx + \int \frac{dx}{x^2+x+\frac{1}{4}+\frac{3}{4}}$$

$$\frac{x^2}{2} - x + \int \frac{dx}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\frac{x^2}{2} - x + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$$

$\therefore$ D is correct answer.

$$\begin{array}{r} x^2+x+1 \overline{) x^3} \\ \underline{x^3+x^2+x} \\ -x^2-2x \\ \underline{-x^2-x-1} \\ x+1 \end{array}$$

3) Since $P(1+2i)=0$ & $P'(1+2i)=0$

$\therefore (1+2i)$ is double root,

Polynomial $P(z)$ has real coefficients

$\therefore (1-2i)$ is also double root.

-3 is real root of $P(z)$

& $P(0) \neq 0$

$\therefore$ Minimum possible degree of $P(z)$ is 5.

Hence **C** is correct answer.

4) $(1+i) = \sqrt{2} \operatorname{Cis} \frac{\pi}{4}$

$$\therefore (1+i)^4 = 4 \operatorname{Cis} \pi$$

$$(1-i) = \sqrt{2} \operatorname{Cis} \left(-\frac{\pi}{4}\right)$$

$$\therefore (1-i)^2 = 2 \operatorname{Cis} \left(-\frac{\pi}{2}\right)$$

$$\begin{aligned} \text{Hence } \frac{(1+i)^4}{(1-i)^2} &= 4 \operatorname{Cis} \pi \div 2 \operatorname{Cis} \left(-\frac{\pi}{2}\right) \\ &= -2i \end{aligned}$$

$\therefore$ New complex number will be twice as far from the origin as P with clockwise rotation by $\frac{\pi}{2}$.

Hence **D** is correct answer.

5) $\underline{a} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} -3 \\ 4 \\ -1 \end{pmatrix}$, $\underline{c} = \begin{pmatrix} 13 \\ 10 \\ 1 \end{pmatrix}$

For perpendicular vectors dot product is zero.

$$\therefore \underline{a} \cdot \underline{b} = 2(-3) - 3(4) + 4(-1)$$

$$= -6 - 12 - 4 = -22$$

$$\underline{b} \cdot \underline{c} = -3(13) + 4(10) - 1(1)$$

$$= -39 + 40 - 1 = 0$$

$$\underline{c} \cdot \underline{a} = 13(2) + 10(-3) + 1(4)$$

$$= 26 - 30 + 4 = 0$$

$\therefore \underline{b}$ and $\underline{a}$ perpendicular to $\underline{c}$.

Hence B is correct answer.

6) The position of the particle for certain values of t :

t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$x = t^2$	0	$\frac{\pi^2}{4}$	π^2	$\frac{9\pi^2}{4}$	$4\pi^2$
$y = 2\cos t$	2	0	-2	0	2
$z = 2\sin t$	0	2	0	-2	0

$\therefore$ particle is rotating in anticlockwise direction in yz plane.

starting from $(2, 0) \rightarrow (0, 2) \rightarrow (-2, 0) \rightarrow (0, -2)$. &

At the same time its x -coordinate is increasing at an increasing rate.

$\therefore$ Motion of the particle is a spiral about the x -axis in an anticlockwise direction.

Hence B is correct answer.

7) P lies on AB in the ratio $AP:PB = 1:2$

$$\vec{AB} = \vec{OB} - \vec{OA} = (-2, -10, 11)$$

$$\therefore \vec{OP} = \vec{OA} + \frac{1}{3}\vec{AB}$$

$$= (4, 3, -5) + \frac{1}{3}(-2, -10, 11)$$

$$= \left(\frac{10}{3}, -\frac{1}{3}, -\frac{4}{3}\right)$$

$\therefore$ A is correct answer.

8) $\int_0^a [f(x-a) + f(a-x)] dx = 0$

$$\text{Let } I = \underbrace{\int_0^a f(x-a) dx}_{I_1} + \underbrace{\int_0^a f(a-x) dx}_{I_2}$$

$$I_1 = \int_0^a f(x-a) dx$$

$$\begin{array}{ll} \text{Let } x-a = u & ; \quad x=0 \Rightarrow u=-a \\ dx = du & \quad x=a \Rightarrow u=0 \end{array}$$

$$\therefore I_1 = \int_{-a}^0 f(u) du$$

$$I_2 = \int_0^a f(a-x) dx$$

$$\begin{array}{ll} \text{Let } a-x = v & ; \quad x=0 \Rightarrow v=a \\ -dx = dv & \quad x=a \Rightarrow v=0 \end{array}$$

$$\therefore I_2 = \int_a^0 -f(v) dv = -\int_a^0 f(v) dv = \int_0^a f(v) dv$$

$$\therefore I = \int_{-a}^0 f(u) du + \int_0^a f(v) dv = \int_{-a}^a f(x) dx$$

Since $I = 0 \quad \forall a > 0$

$$\therefore \int_{-a}^a f(x) dx = 0 \quad \forall a > 0$$

$\Rightarrow f(x)$ is odd function.

- Integral of an odd function over a symmetric interval $[-a, a]$ is zero
- Conversely, if the integral over every symmetric interval is zero, the function must be odd.

$\therefore$ D is correct answer.

9) Since $\frac{dv}{dx} = \frac{2}{3v}$

$$\therefore \text{acceleration} = v \frac{dv}{dx} = v \cdot \frac{2}{3v} = \frac{2}{3} \text{ (constant)}$$

acceleration is constant.

For velocity: $\frac{dv}{dt} = \frac{2}{3}$

$$dv = \frac{2}{3} dt$$

$$v = \frac{2}{3}t + C$$

The particle starts from rest at the origin

$$\therefore \text{when } t=0, v=0 \text{ then } C=0$$

$$\therefore v = \frac{2}{3}t$$

As t increases, v increases

$\therefore$ The particle has constant acceleration and increasing velocity.

Hence C is correct answer.

$$10) \quad \omega^2 + \frac{1}{1+\omega^2} + \omega = 0$$

$$\omega^2(1+\omega^2) + 1 + \omega(1+\omega^2) = 0$$

$$\omega^2 + \omega^4 + 1 + \omega + \omega^3 = 0$$

$$\omega^4 - \omega^3 + \omega^2 - \omega + 1 = 0$$

$$\frac{\omega^5 - 1}{\omega - 1} = 0$$

$$\omega^5 - 1 = 0 \quad (\because \omega \neq -1)$$

$$\therefore \omega^5 = 1$$

$$\omega^{2026} = (\omega^5)^{405} \cdot \omega$$

$$= (1)^{405} \cdot \omega$$

$$= \omega$$

$\therefore$ A is correct answer.

END OF SECTION I

Section II

Question-11

a) $z = 1 + ai$, $w = a + i$, $a \in \mathbb{R}$

i) $z + \bar{w} = 1 + ai + (\overline{a+i})$

1 Correct Solution

$$= 1 + ai + a - i$$

$$= (1+a) + i(a-1)$$

ii) $zw = (1+ai)(a+i)$

2 Correct Solution

$$= a + i + a^2i + ai^2$$

1 Find expression for zw .

$$= a + i + a^2i - a$$

$$= i(1+a^2)$$

Since $a^2+1 > 0$, and zw is purely imaginary,

$$\therefore \text{Arg}(zw) = \frac{\pi}{2}$$

iii) $\frac{z}{w} = \frac{1+ai}{a+i}$

$$= \frac{1+ai}{a+i} \times \frac{(a-i)}{(a-i)}$$

$$= \frac{(a-a^2) + i(a^2-1)}{a^2+1}$$

$$= \frac{2a + i(a^2-1)}{a^2+1}$$

For purely real; $\text{Im}\left(\frac{z}{w}\right) = 0$

2 Correct Solution

$$\therefore \frac{a^2-1}{a^2+1} = 0$$

1 Recognise $\text{Im}\left(\frac{z}{w}\right) = 0$

$$a^2 - 1 = 0$$

$$a^2 = 1$$

$$a = \pm 1$$

$$11b) \quad \underline{a} + \underline{b} = \underline{i} \quad , \quad \underline{b} = 3\underline{i} + 2\underline{j} - 2\underline{k}$$

$$\underline{a} = \underline{i} - \underline{b}$$

$$= \underline{i} - 3\underline{i} - 2\underline{j} + 2\underline{k}$$

$$= -2\underline{i} - 2\underline{j} + 2\underline{k}$$

$$|\underline{a}| = \sqrt{(-2)^2 + (-2)^2 + (2)^2}$$

$$= \sqrt{4 + 4 + 4}$$

$$= 2\sqrt{3}$$

2 Correct Solution

1 Correctly find a .

$$11c) \quad \int \frac{x^3}{x^2+9} dx$$

$$\int \frac{x^3}{x^2+9} dx = \int \frac{x(x^2+9) - 9x}{x^2+9} dx$$

$$= \int \left(x - \frac{9x}{x^2+9} \right) dx$$

$$= \int x dx - \frac{9}{2} \int \frac{2x}{x^2+9} dx$$

$$= \frac{x^2}{2} - \frac{9}{2} \ln(x^2+9) + C$$

2 Correct Solution

1 Correctly express the given integral in terms of u by suitable substitution or equivalent merit.

11d) i) If n is not odd then $n^2 - 6n + 5$ is not even. 1 correct solution.

ii) Let $n = 2k$, $k \in \mathbb{Z}$

$$\text{then } n^2 - 6n + 5 = (2k)^2 - 6(2k) + 5$$

2 Correct solution.

$$= 4k^2 - 12k + 5$$

1 Use proof b

$$= 2(2k^2 - 6k + 2) + 1$$

Contrapositive or
equivalent merit.

$$= 2M + 1 \quad (\text{Since } k \in \mathbb{Z} \therefore M \in \mathbb{Z})$$

$\therefore n^2 - 6n + 5$ is odd, if n is even, and $M = 2k^2 - 6k + 2 \in \mathbb{Z}$

Hence, P has been proven by contraposition.

11e) $z^2 - z + (4 - 2i) = 0$

By quadratic formula:

3 Correct solution

$$z = \frac{1 \pm \sqrt{1 - 4(4 - 2i)}}{2}$$

2 Uses quadratic formula

$$= \frac{1 \pm \sqrt{1 - 16 + 8i}}{2}$$

$$\text{and } \sqrt{-15 + 8i} = \pm(1 + 4i)$$

$$= \frac{1 \pm \sqrt{-15 + 8i}}{2}$$

1 Uses quadratic formula

To find sq. roots of $-15 + 8i$:

or finds the $\Delta = -15 + 8i$

$$\text{Let } (a + ib)^2 = -15 + 8i$$

$$a^2 - b^2 = -15 \text{ \& } 2ab = 8$$

Solve simultaneously; $a = \pm 1$ & $b = \pm 4$

$$\therefore \sqrt{-15 + 8i} = \pm(1 + 4i)$$

$$\therefore z = \frac{1 \pm (1 + 4i)}{2}$$

$$= \frac{2 + 4i}{2}, \frac{-4i}{2}$$

$$= 1 + 2i, -2i$$

Question-12

$$a) P(z) = z^4 - 8z^3 + pz^2 + qz - 80$$

As the coefficients of this polynomial are real, the conjugate of the root $(3+i)$, which is $(3-i)$ is also a root.

Let the roots of $P(z)$ be:

$$(3+i), (3-i), \alpha, \beta$$

$$\text{Sum of roots: } (3+i) + (3-i) + \alpha + \beta = 8$$

$$6 + \alpha + \beta = 8$$

$$\alpha + \beta = 2 \quad \text{--- (1)}$$

$$\text{Product of roots: } (3+i)(3-i)\alpha\beta = -80$$

$$10\alpha\beta = -80$$

$$\alpha\beta = -8 \quad \text{--- (2)}$$

3 Correct Solution.

The quad. eq. with α and β is

$$z^2 - 2z - 8 = 0$$

$$(z-4)(z+2) = 0$$

$$\therefore z = 4 \text{ or } -2$$

2 Significant progress to find all four roots.

1 Recognise $(3-i)$ as second root.

Hence four roots are: $3+i, 3-i, 4, -2$

$$ii) P(z) = (z-3-i)(z-3-i)(z-4)(z+2)$$

$$= ((z-3)^2 - i^2)(z^2 - 2z - 8)$$

1 Correct Solution.

$$= (z^2 - 6z + 9 + 1)(z^2 - 2z - 8)$$

$$= (z^2 - 6z + 10)(z^2 - 2z - 8)$$

$$b) \quad a^2 + b^2 = (a-b)^2 + 2ab$$

$$> (a-b)^2 + 2 \quad (\because ab > 1)$$

$$\frac{a^2 + b^2}{a-b} \geq (a-b) + \frac{2}{(a-b)} \quad (\because a > b)$$

$$\geq 2\sqrt{(a-b)\frac{2}{(a-b)}} \quad (\because \text{AM/GM})$$

$$= 2\sqrt{2}$$

Hence proved.

2 Correct Solution.

1 Significant progress towards proof by using given conditions $a > b$ and $ab > 1$ or AM/GM.

$$c) \quad \text{Since } l_1 \perp l_2$$

$\therefore$ Dot product of their direction vector is zero.

$$\text{i.e.} \quad \begin{pmatrix} 2 \\ \ln a \\ \ln a \end{pmatrix} \cdot \begin{pmatrix} -3 \\ \ln a \\ 1 \end{pmatrix} = 0$$

$$-6 + (\ln a)^2 + \ln a = 0$$

$$(\ln a)^2 + (\ln a) - 6 = 0$$

$$(\ln a + 3)(\ln a - 2) = 0$$

$$\ln a = -3 \quad \text{or} \quad \ln a = 2$$

$$a = e^{-3} \quad \text{or} \quad a = e^2.$$

3 Correct Solution

2 Solve the quadratic equation in $\ln a$.

1 Use dot product

$$12d) \int_0^{\pi/4} \frac{4}{5-3\cos\theta} d\theta$$

$$\text{Let } t = \tan\theta$$

$$dt = \sec^2\theta d\theta$$

$$dt = (1 + \tan^2\theta) d\theta$$

$$dt = (1 + t^2) d\theta$$

$$\text{At } \theta = 0, t = 0$$

$$\text{At } \theta = \frac{\pi}{4}, t = 1$$

$$d\theta = \frac{dt}{1+t^2}$$

$$\therefore \int_0^{\pi/4} \frac{4}{5-3\cos\theta} d\theta = \int_0^1 \frac{4}{5-3\frac{(1-t^2)}{(1+t^2)}} \cdot \frac{dt}{(1+t^2)}$$

$$= \int_0^1 \frac{4 dt}{5(1+t^2) - 3(1-t^2)}$$

$$= \int_0^1 \frac{4 dt}{2 + 8t^2}$$

$$= \int_0^1 \frac{2 dt}{1 + 4t^2}$$

$$= [\tan^{-1}(2t)]_0^1$$

$$= \tan^{-1}2 - \tan^{-1}0$$

$$= \tan^{-1}2$$

3 Correct Solution.

2 Finds Primitive function

1 Express integrand in terms of t.

$$12e) i) x = 5 - 4 \cos^2 2\pi t$$

$$x = 5 - 2(2 \cos^2 2\pi t)$$

$$= 5 - 2(1 + \cos 4\pi t) \quad (\because 2 \cos^2 \theta = 1 + \cos 2\theta)$$

$$= 3 - 2 \cos 4\pi t$$

2 Correct Solution

$$\therefore \text{Centre of motion} = 3$$

1 Correct centre of motion

$$\text{Amplitude} = 2$$

or amplitude or express

$$x = 3 - 2 \cos 4\pi t$$

$$ii) \text{ Period } (T) = \frac{2\pi}{4\pi} = \frac{1}{2}$$

1 Correct solution.

$$\text{Since Amplitude} = 2 \text{ cm}$$

$$\therefore \text{In every } \frac{1}{2} \text{ second particle travel } 8 \text{ cm}$$

$$\therefore \text{In 5 seconds particle will travel } 8 \times 2 \times 5 = 80 \text{ cm.}$$

Question-13

$$a) P(x) = 120 + 120x + 60x^2 + 20x^3 + 5x^4 + x^5$$

By Contradiction:

Let $P(x)$ has double root at $x = \alpha$

$$\therefore P(\alpha) = 0 \quad \text{and} \quad P'(\alpha) = 0$$

$$P'(x) = 120 + 120x + 60x^2 + 20x^3 + 5x^4$$

$$P(x) - P'(x) = x^5$$

$$\therefore P(\alpha) - P'(\alpha) = \alpha^5$$

$$\text{Since } P(\alpha) = P'(\alpha) = 0$$

$$\therefore \alpha^5 = 0$$

$$\Rightarrow \alpha = 0$$

$\therefore$ Double root is at $x = 0$

$$\text{However } P(0) = 120 \neq 0$$

which is a contradiction.

Hence $P(x)$ has no double root.

2 Correct Solution.

1 Obtains $P(x)$ has double root at $x = \alpha$ &

$$P(\alpha) = 0 = P'(\alpha)$$

$$b) \quad I_n = \int_0^1 \frac{dx}{(1+x^2)^n}, \quad n \geq 0$$

$$\text{Let } u = (1+x^2)^{-n} \quad dv = dx$$

$$du = -2nx(1+x^2)^{-n-1} \quad v = x$$

$$\therefore I_n = \left[x(1+x^2)^{-n} \right]_0^1 + 2n \int_0^1 x^2(1+x^2)^{-n-1} dx$$

$$= [1(1+1)^{-n} - 0] + 2n \int_0^1 \frac{x^2 + 1 - 1}{(1+x^2)^{n+1}} dx$$

$$= 2^{-n} + 2n \int_0^1 \left(\frac{x^2+1}{(1+x^2)^{n+1}} - \frac{1}{(1+x^2)^{n+1}} \right) dx$$

$$= 2^{-n} + 2n \int_0^1 \left(\frac{1}{(1+x^2)^n} - \frac{1}{(1+x^2)^{n+1}} \right) dx$$

$$= \frac{1}{2^n} + 2n(I_n - I_{n+1})$$

$$= \frac{1}{2^n} + 2nI_n - 2nI_{n+1}$$

$$2nI_{n+1} = \frac{1}{2^n} + 2nI_n - I_n$$

$$2nI_{n+1} = \frac{1}{2^n} + (2n-1)I_n$$

$$I_{n+1} = \frac{1}{2n} \left[\frac{1}{2^n} + (2n-1)I_n \right]$$

Hence shown.

3 Correct Solution

2 Uses an appropriate technique to find I_n or I_{n+1} after applying integration by parts

1 Applies integration by parts

ii) Sub $n=2$ in part i)

$$I_3 = \frac{1}{4} \left[\frac{1}{4} + 3I_2 \right]$$

$$= \frac{1}{16} + \frac{3}{4}I_2$$

$$I_2 = \frac{1}{2} \left[\frac{1}{2} + I_1 \right]$$

$$= \frac{1}{4} + \frac{1}{2}I_1$$

$$I_1 = \int_0^1 \frac{dx}{1+x^2} = \left[\tan^{-1}x \right]_0^1$$
$$= \tan^{-1}1 - \tan^{-1}0$$

$$= \frac{\pi}{4}$$

$$\therefore I_2 = \frac{1}{4} + \frac{\pi}{8}$$

$$\& I_3 = \frac{1}{16} + \frac{3}{4} \left(\frac{1}{4} + \frac{\pi}{8} \right)$$

$$= \frac{1}{16} + \frac{3}{16} + \frac{3\pi}{32}$$

$$= \frac{1}{4} + \frac{3\pi}{32}$$

c) i) Mass of particle = 1 kg.

Forces are :

$$\text{Gravitation force } \underline{F_g} = (0, -10)$$

$$\text{Air resistance} = -5\underline{v}$$

$$\text{Hence } m\underline{a} = m\underline{F_g} - 5\underline{v}$$

$$\underline{a} = \begin{pmatrix} 0 \\ -10 \end{pmatrix} - 5\underline{v} \quad (\because m=1)$$

$$\therefore \begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = \begin{pmatrix} -5\dot{x} \\ -10 - 5\dot{y} \end{pmatrix}$$

2 Correct solution.

1 Attempts to use previous

result as a recurrence

formula to find I_2 or I_3

or find $I_1 = \frac{\pi}{4}$

Horizontal Component:

$$\ddot{x} = -5x$$

$$\text{or } \frac{dx}{dt} = -5x$$

$$\int \frac{dx}{x} = -5 \int dt$$

$$\ln x = -5t + C$$

$$x = e^{-5t+C}$$

$$x = A e^{-5t} \quad (A = e^C)$$

$$\text{At } t=0, \quad \dot{x} = V \cos \alpha$$

$$\therefore A = V \cos \alpha$$

$$\text{Hence } \dot{x} = V e^{-5t} \cos \alpha$$

Vertical Component

$$\ddot{y} = -5\dot{y} - 10$$

$$\frac{d\dot{y}}{dt} = -5(\dot{y} + 2)$$

$$\frac{d\dot{y}}{\dot{y} + 2} = -5 dt$$

$$\int \frac{d\dot{y}}{\dot{y} + 2} = -5 \int dt$$

$$\ln(\dot{y} + 2) = -5t + C$$

$$\dot{y} + 2 = e^{-5t+C}$$

$$\dot{y} = B e^{-5t} - 2 \quad (B = e^C)$$

$$\text{At } t=0, \quad \dot{y} = V \sin \alpha$$

$$\therefore B = V \sin \alpha + 2$$

$$\text{Hence } \dot{y} = (V \sin \alpha + 2) e^{-5t} - 2$$

$$\text{Hence } \underline{v}(t) = \begin{pmatrix} V e^{-5t} \cos \alpha \\ (V \sin \alpha + 2) e^{-5t} - 2 \end{pmatrix}$$

3 Correct Solution

2 Correctly find $\dot{x}$ or $\dot{y}$

1 Express $a = F_g - 5V$ or $\ddot{x} = -5x$ &

$$\ddot{y} = -5\dot{y} - 10$$

ii) Sub $t=1$ in part i)

$$\therefore \underline{v}(1) = \begin{pmatrix} V e^{-5} \cos \alpha \\ (V \sin \alpha + 2) e^{-5} - 2 \end{pmatrix} \quad \text{--- ①}$$

$$\text{However, given that } \underline{v}(1) = \begin{pmatrix} 250 e^{-5} \\ (250\sqrt{3} + 2) e^{-5} - 2 \end{pmatrix} \quad \text{--- ②}$$

$$\text{Equate ① \& ② ; } V \cos \alpha = 250 \quad \& \quad V \sin \alpha = 250\sqrt{3}$$

$$\therefore V^2 = (250)^2 + (250\sqrt{3})^2 \Rightarrow V = 500 \text{ m/s}$$

$$\& \tan \alpha = \sqrt{3} \Rightarrow \alpha = 60^\circ$$

2 Correct solution

1 Correct value for V
or α or finding
two equation

$$V \cos \alpha = 250$$

$$\& V \sin \alpha = 250\sqrt{3}$$

iii) Sub $V = 500$ and $\alpha = 60^\circ$ in $\underline{v}(t)$

$$\begin{aligned} \underline{v}(t) &= \begin{pmatrix} 500 e^{-5t} \cos 60^\circ \\ (500 \sin 60^\circ + 2) e^{-5t} - 2 \end{pmatrix} \\ &= \begin{pmatrix} 250 e^{-5t} \\ (250\sqrt{3} + 2) e^{-5t} - 2 \end{pmatrix} \end{aligned}$$

At $t=0$; horizontal velocity is $\dot{x} = 250$

let at time $t=T$, the particle reaches at max. height.

$$\therefore \text{At } t=T; \quad \dot{y} = 0$$

$$\therefore (250\sqrt{3} + 2) e^{-5T} - 2 = 0$$

$$e^{-5T} = \frac{2}{250\sqrt{3} + 2}$$

$$= \frac{1}{125\sqrt{3} + 1}$$

$\therefore$ At $t = T$; Horizontal velocity is $\dot{x} = 250 e^{-5T}$

$$\therefore \dot{x} = \frac{250}{125\sqrt{3} + 1}$$

Ratio of two horizontal velocities is

$$\begin{aligned} 250 &: \frac{250}{125\sqrt{3} + 1} \\ &= (125\sqrt{3} + 1) : 1 \end{aligned}$$

Hence shown

2 Correct solution

1 Find time at max height or horizontal velocity at max. height.

Question 14

a) Since $a, b, c \in \mathbb{R}^+$

$$\therefore \frac{a+b+c}{3} \geq \sqrt[3]{abc} \quad (\text{AM/GM Property})$$

$$\therefore \frac{1}{3} \geq \sqrt[3]{abc} \quad (\because a+b+c=1) \quad \text{--- (1)}$$

Also, for $a, b, c > 0$

$$\therefore \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \sqrt[3]{\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3 \left(\sqrt[3]{\frac{1}{abc}} \right)$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3 (3) \quad \text{--- (from (1) } \frac{1}{\sqrt[3]{abc}} \geq 3)$$

$$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$$

Hence proved.

2 Correct solution.

1 Uses AM/GM along with substitution of $a+b+c=1$

$$\begin{aligned}
\text{ii)} \quad \text{L.H.S} &= \left(\frac{1}{a} - 1\right) \left(\frac{1}{b} - 1\right) \left(\frac{1}{c} - 1\right) \\
&= \left(\frac{1-a}{a}\right) \left(\frac{1-b}{b}\right) \left(\frac{1-c}{c}\right) \\
&= \frac{(1-a)(1-b)(1-c)}{abc} \\
&= \frac{1 - (a+b+c) + (ab+bc+ca) - abc}{abc} \\
&= \frac{1 - 1 + ab + bc + ca - abc}{abc} \quad (\because a+b+c=1) \\
&= \frac{ab + bc + ca - abc}{abc} \\
&= \frac{ab}{abc} + \frac{bc}{abc} + \frac{ca}{abc} - \frac{abc}{abc} \\
&= \frac{1}{c} + \frac{1}{a} + \frac{1}{b} - 1 \\
&\geq 9 - 1 \quad (\text{By part i}) \\
&= 8 \\
&= \text{R.H.S}
\end{aligned}$$

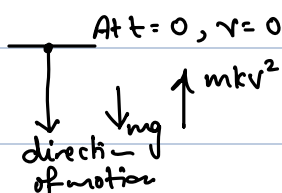
Hence proved.

3 Correct Solution.

2 Significant progress towards the solution using the result of part i)

1 Correctly expand L.H.S $\left(\frac{1}{a}-1\right)\left(\frac{1}{b}-1\right)\left(\frac{1}{c}-1\right)$ or equivalent merit

14b)



Equation of motion:

$$m\ddot{x} = mg - mkv^2$$

$$\therefore \ddot{x} = g - kv^2$$

$$v \frac{dv}{dx} = g - kv^2$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$x = -\frac{1}{2k} \ln(g - kv^2) + C$$

$$\text{At } x=0, v=0 \Rightarrow 0 = -\frac{1}{2k} \ln g + C$$

$$\therefore x = \frac{1}{2k} \ln g - \frac{1}{2k} \ln(g - kv^2)$$

$$= \frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right)$$

$$2kx = \ln\left(\frac{g}{g - kv^2}\right)$$

$$e^{2kx} = \frac{g}{g - kv^2}$$

$$g - kv^2 = g e^{-2kx}$$

$$kv^2 = g - g e^{-2kx}$$

$$v^2 = \frac{g}{k} (1 - e^{-2kx}) \quad \text{--- (1)}$$

ii) As $x \rightarrow \infty, v^2 \rightarrow \frac{g}{k} \quad (\because e^{-2kx} \rightarrow 0)$

$$\therefore v_T = \sqrt{\frac{g}{k}}$$

3 Correct solution

2 Finds x as a function of v .

1 Correct expression for the resultant force.

1 Correct solution

iii) When $v = \frac{1}{2} v_T = \frac{1}{2} \sqrt{\frac{g}{k}}$

$$v^2 = \frac{g}{4k}$$

Sub in ① $\frac{g}{4k} = \frac{g}{k} (1 - e^{-2kx})$

$$\frac{1}{4} = 1 - e^{-2kx}$$

$$e^{-2kx} = 1 - \frac{1}{4}$$

$$e^{-2kx} = \frac{3}{4}$$

$$e^{2kx} = \frac{4}{3}$$

$$2kx = \ln \frac{4}{3}$$

$$x = \frac{1}{2k} \ln \frac{4}{3}$$

Hence Shown.

2 Correct solution

1 Sub. $v = \frac{1}{2} v_T$ in the expression of part i) or equivalent merit

$$\begin{aligned} \text{C) i) L.H.S} &= \frac{1}{x \sqrt{x^{2n}-1}} = \frac{1}{x \sqrt{x^{2n}(1-x^{-2n})}} \\ &= \frac{1}{x \cdot x^n \sqrt{1-x^{-2n}}} \\ &= \frac{x^{-1-n}}{\sqrt{1-x^{-2n}}} \\ &= \text{R.H.S} \end{aligned}$$

1 Correct Solution

c ii)

$$\int_2^4 \frac{1}{x \sqrt{x^{2n}-1}} dx = \int_2^4 \frac{x^{-n-1}}{\sqrt{1-(x^{-n})^2}} dx$$

Let $u = x^{-n}$; $x=2 \Rightarrow u=2^{-n}$

$du = -n x^{-n-1} dx$ $x=4 \Rightarrow u=4^{-n}$

$$= -\frac{1}{n} \int_{2^{-n}}^{4^{-n}} \frac{1}{\sqrt{1-u^2}} du$$

3 Correct solution.

2 Finds the primitive function

$$= \frac{1}{n} \left[\cos^{-1} u \right]_{2^{-n}}^{4^{-n}}$$

1 Uses part i) to rearrange integrand and substitute $u = x^{-n}$ or equivalent merit

$$= \frac{1}{n} \left[\cos^{-1}(4^{-n}) - \cos^{-1}(2^{-n}) \right]$$

or $-\frac{1}{n} \left[\sin^{-1}(4^{-n}) - \sin^{-1}(2^{-n}) \right]$

iii)

$$\int_2^4 \frac{dx}{x \sqrt{x^{2n}-1}} > \int_2^4 \frac{dx}{x \sqrt{x^{2n}}} \quad (\because x^{2n} > x^{2n}-1)$$

$$= \int_2^4 \frac{dx}{x^{n+1}} = \int_2^4 x^{-(n+1)} dx$$

$$= -\frac{1}{n} \left[x^{-n} \right]_2^4$$

$$= -\frac{1}{n} \left[4^{-n} - 2^{-n} \right]$$

$$= \frac{1}{n} \left[2^{-n} - 4^{-n} \right]$$

$$= \frac{1}{n} \left[\frac{1}{2^n} - \frac{1}{4^n} \right]$$

$$= \frac{1}{n} \left[\frac{2^n - 1}{4^n} \right]$$

Hence shown

2 Correct solution

1 Identifies $x^{2n} > x^{2n}-1$

or equivalent merit.

Question 15

a) $\ddot{x} = 4x^3 - 16x$

$$x \frac{dx}{dt} = 4x^3 - 16x$$

$$\int_{v_0}^{\dot{x}} x dx = \int_0^x (4x^3 - 16x) dx$$

$$\left[\frac{\dot{x}^2}{2} \right]_{v_0}^{\dot{x}} = \left[x^4 - 8x^2 \right]_0^x$$

$$\frac{1}{2} [\dot{x}^2 - v_0^2] = x^4 - 8x^2$$

$$\dot{x}^2 - v_0^2 = 2x^4 - 16x^2$$

$$\dot{x}^2 - 2x^4 + 16x^2 = v_0^2$$

2 correct solution

1 uses $x \frac{dx}{dt}$ or

equivalent merit.

Since v_0^2 is constant, therefore $(\dot{x})^2 - 2x^4 + 16x^2$ does not change with time.

Alternatively:

To show $(\dot{x})^2 - 2x^4 + 16x^2$ does not change with time, we

can show $(\dot{x})^2 - 2x^4 + 16x^2 = \text{constant}$

$$\frac{d}{dt} [(\dot{x})^2 - 2x^4 + 16x^2] = 2\dot{x}\ddot{x} - 8x^3\dot{x} + 32x\dot{x}$$

$$= 2\dot{x}[\ddot{x} - 4x^3 + 16x]$$

$$= 2\dot{x}[\ddot{x} - \ddot{x}] \quad (\because \ddot{x} = 4x^3 - 16x)$$

$$= 0$$

2 correct solution

$$\text{Since } \frac{d}{dt} [\dot{x}^2 - 2x^4 + 16x^2] = 0$$

$$\therefore \dot{x}^2 - 2x^4 + 16x^2 = \text{constant.}$$

1 Correctly differentiate the given quantity w.r.t. time.

a) ii) When $r_0 = \sqrt{\frac{31}{8}}$, $(\dot{x})^2 - 2x^4 + 16x^2 = \frac{31}{8}$

2 correct solution.

$$(\dot{x})^2 = 2x^4 - 16x^2 + \frac{31}{8}$$

1 Uses the fact that

$$= \frac{1}{8} (16x^4 - 128x^2 + 31)$$

$(\dot{x})^2 \geq 0$ to find the

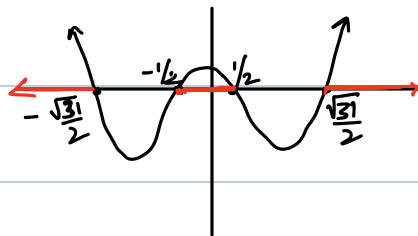
$$= \frac{1}{8} (4x^2 - 31)(4x^2 - 1)$$

region.

Since $(\dot{x})^2 \geq 0$

$$\therefore (4x^2 - 31)(4x^2 - 1) \geq 0$$

$$(2x - \sqrt{31})(2x + \sqrt{31})(2x - 1)(2x + 1) \geq 0$$



However as $x=0$ when $t=0$, therefore particle must remain in the region $-\frac{1}{2} \leq x \leq \frac{1}{2}$.

a) iii) $\ddot{x} = 4x^3 - 16x^2$

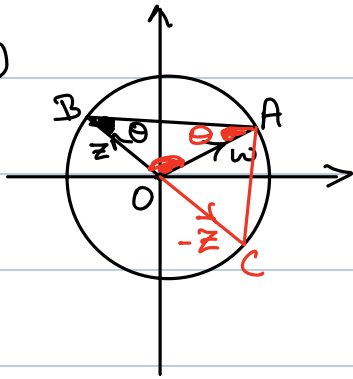
1 correct solution.

$$= 4x^2(x - 4), \text{ which is not in the form of } \ddot{x} = -n^2x$$

$\therefore$ The particle is not moving in simple harmonic motion.

15b)

i)



In $\triangle OAB$

Since $\angle OBA = \theta$

$$\therefore \angle OAB = \theta \quad (\because OA = OB)$$

$$\therefore \angle BOA = \pi - 2\theta \quad (\text{Angle sum of } \triangle OAB)$$

$$\therefore z = w \operatorname{cis}(\pi - 2\theta)$$

$$= w [\cos(\pi - 2\theta) + i \sin(\pi - 2\theta)]$$

$$= w (-\cos 2\theta + i \sin 2\theta)$$

Hence shown.

2 Correct Solution.

1 Find $\angle BOA$ or equivalent merit.

$$ii) \vec{AB} = z - w$$

$$\vec{OC} = -z$$

$$\vec{CA} = \vec{OA} - \vec{OC}$$

$$= w + z$$

In $\triangle ABC$; $\angle BAC = \frac{\pi}{2}$ (Angle in semicircle)

$$\therefore \tan \theta = \frac{|\vec{AC}|}{|\vec{AB}|}$$

$$= \frac{|w + z|}{|z - w|}$$

$$\therefore |z + w| = \tan \theta |z - w|$$

2 Correct Solution

1 Find $\vec{AB}$ or $\vec{AC}$ or

$\angle BAC = \frac{\pi}{2}$ or equivalent

merit.

The diagram illustrates a 3D geometric figure, possibly a prism or a pyramid, with vertices labeled A, B, C, D, E, F, and X. A red line segment connects vertex B to vertex X. A dashed line segment connects vertex A to vertex F. A vertical dashed line segment connects vertex D to vertex X. Tick marks are present on segments FX and XE, indicating that FX = XE.

i) To show $\vec{XD} \perp \vec{EF}$, prove that $\vec{XD} \cdot \vec{EF} = 0$

3 Correct Solution

2 Applies the property of

equilateral Δ to the dot product.

or equivalent merit.

1 Express $\vec{XD} = \vec{XF} + \vec{FD}$ or attempt

to use dot product $\vec{x} \cdot \vec{F}$.

2

($\because \triangle DEF$ is equilateral
 $\therefore \angle DFE = 60^\circ$)

ii) Join B to X; In ΔBFX

2 Correct Solution

1 Use triangle inequality

or equivalent merit.

$$= \frac{1}{2}m + 2n$$

$= 2(m+n)$, which is perimeter of rectangle BCFG.

Hence shown.

Question 16

$$a) \int_0^{\frac{\pi}{4}} \frac{1}{\cos x \sqrt{\sec x - \tan x}} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sec x}{\sqrt{\sec x - \tan x}} dx$$

$$\text{Let } \sec x - \tan x = u$$

$$\text{At } x=0 \Rightarrow u = \sec 0 - \tan 0 = 1$$

$$\text{At } x = \frac{\pi}{4} \Rightarrow u = \sec \frac{\pi}{4} - \tan \frac{\pi}{4} = \sqrt{2} - 1$$

$$(\sec x \tan x - \sec^2 x) dx = du$$

$$\sec x (\tan x - \sec x) dx = du$$

$$\sec x dx = \frac{du}{-u}$$

$$\therefore \int_0^{\frac{\pi}{4}} \frac{\sec x}{\sqrt{\sec x - \tan x}} dx = \int_1^{\sqrt{2}-1} \frac{1}{\sqrt{u}} \left(-\frac{du}{u} \right)$$

$$= \int_1^{\sqrt{2}-1} -u^{-3/2} du$$

$$= 2 \left[u^{-1/2} \right]_1^{\sqrt{2}-1}$$

$$= 2 \left[\frac{1}{\sqrt{\sqrt{2}-1}} - 1 \right]$$

$$= \frac{2}{\sqrt{2}-1} - 2$$

4 Correct solution.

3 Correctly integrate after substitution.

2 Express x in terms of u along with both limits.

1 Express x in terms of u after substitution.

16 b) $A(-5, -4)$, Let $B(b, 5)$

and $C(c, -12)$

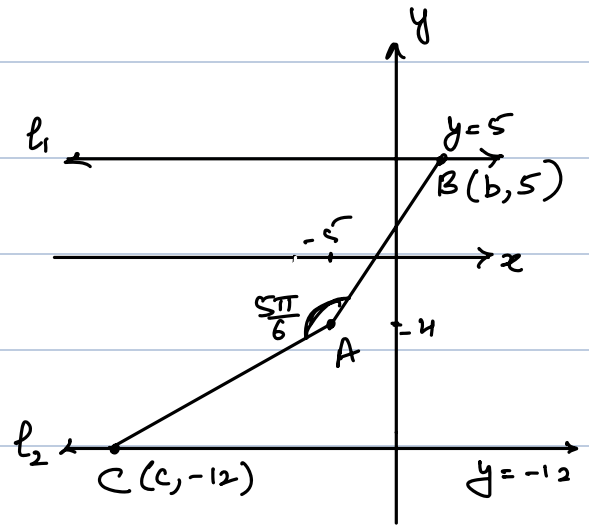
$$\vec{AC} = 2\vec{AB} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

$$= 2\vec{AB} \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right)$$

$$\vec{AC} = \vec{AB}(-\sqrt{3} + i)$$

$$\vec{AC} = (c+5) - 8i$$

$$\vec{AB} = (b+5) + 9i$$



$$\begin{aligned} \therefore (c+5) - 8i &= [(b+5) + 9i](-\sqrt{3} + i) \\ &= [-\sqrt{3}(b+5) - 9] + i[(b+5) - 9\sqrt{3}] \end{aligned}$$

Compare real and imaginary parts;

$$-8 = (b+5) - 9\sqrt{3} \Rightarrow b = 9\sqrt{3} - 13$$

$$c+5 = -\sqrt{3}(b+5) - 9$$

$$c = -\sqrt{3}(9\sqrt{3} - 13 + 5) - 9 - 5$$

$$= -27 + 8\sqrt{3} - 14$$

$$= 8\sqrt{3} - 41$$

$$\therefore z_c = 8\sqrt{3} - 41 - 12i$$

3 Correct Solution

2 Equating real and imaginary part
to find $b = 9\sqrt{3} - 13$ or significant
progress towards solution.

1 Express $\vec{AC} = 2\vec{AB} \text{cis} \frac{5\pi}{6}$

16 c) i) Let Statement $S(r)$ be $z_r = i^r \binom{n}{r} z_0$

Prove $S(0)$ is true ; let $r=0$

$$\text{L.H.S} = z_0$$

$$\text{R.H.S} = i^0 \binom{n}{0} z_0$$

$$= 1(z_0)$$

$$= z_0$$

$\therefore S(0)$ is true.

Assume $S(p)$ is true for $p \geq 0$

$$\text{i.e } z_p = i^p \binom{n}{p} z_0$$

————— ①

Prove $S(p+1)$ is true

$$\text{i.e } z_{p+1} = i^{p+1} \binom{n}{p+1} z_0$$

Sub $z=p$ in given condition

$$\therefore (p+1)z_{p+1} - i(n-p)z_p = 0$$

$$(p+1)z_{p+1} = i(n-p)z_p$$

$$z_{p+1} = i \left(\frac{n-p}{p+1} \right) z_p$$

$$= i \left(\frac{n-p}{p+1} \right) \cdot i^p \binom{n}{p} z_0 \quad (\text{Using } ①)$$

$$= i^{p+1} \left(\frac{n-p}{p+1} \right) \frac{n!}{p!(n-p)!} z_0$$

$$= i^{p+1} \frac{n!}{(p+1)!(n-p-1)!} z_0$$

$$= i^{p+1} \binom{n}{p+1} z_0$$

Hence true for $S(p+1)$ if true for $S(p)$

$\therefore S(p) \text{ true} \Rightarrow S(p+1) \text{ true}$ and $S(0)$ is true. Therefore by principle of mathematical induction $S(r)$ is true for all integers $r \geq 0$.

4 key points

1. Proving true for $n=0$
2. Clearly stating the assumption and what to be proved.
3. Using the assumption in the proof.
4. Correctly proving the required result

3 Correctly done all 4 key points.

2 Correctly done 3 out of 4 key points.

1 Correctly done 2 out of 4 key points.

16 c) ii) Using the given identity:

$$Z_0 + Z_1 + Z_2 + \dots + Z_n = 2^n$$

$$Z_0 + i \binom{n}{1} Z_0 + i^2 \binom{n}{2} Z_0 + \dots + i^n \binom{n}{n} Z_0 = 2^n$$

(Using the result of part (i))

$$Z_0 \left[1 + i \binom{n}{1} + i^2 \binom{n}{2} + \dots + i^n \binom{n}{n} \right] = 2^n$$

$$Z_0 (1+i)^n = 2^n$$

$$\begin{aligned} Z_0 &= \left(\frac{2}{1+i} \right)^n \\ &= \left(\frac{2}{1+i} \times \frac{1-i}{1-i} \right)^n \\ &= \left[\frac{2(1-i)}{2} \right]^n \\ &= (1-i)^n \end{aligned}$$

2 Correct solution.

1 Uses result of part i) to get $Z_0(1+i)^n = 2^n$

Hence shown.

c) iii) $|Z_0|^2 + |Z_1|^2 + |Z_2|^2 + \dots + |Z_n|^2$

$$= |Z_0|^2 + |i \binom{n}{1} Z_0|^2 + |i^2 \binom{n}{2} Z_0|^2 + \dots + |i^n \binom{n}{n} Z_0|^2$$

$$= |Z_0|^2 \left[\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \dots + \binom{n}{n}^2 \right]$$

$$= |(1-i)^n|^2 \binom{2n}{n} \quad \left(\because \text{Given that } \sum_{m=0}^n \binom{n}{m}^2 = \binom{2n}{n} \right)$$

$$= 2^n \binom{2n}{n}$$

$$= 2^n \frac{(2n)!}{n! n!}$$

$$= \frac{2^n}{n!} (2n)(2n-1)(2n-2) \dots (n+1)$$

$$\leq \frac{2^n}{n!} \left[\frac{2n + (2n-1) + (2n-2) + \dots + (n+1)}{n} \right]^n \quad (\because \text{AM/GM inequality})$$

$$= \frac{2^n}{n!} \left(\frac{3n+1}{2} \right)^n \quad (\text{Taking sum of A.P.})$$

$$= \frac{(3n+1)^n}{n!}$$

$$2n + (2n-1) + (2n-2) + \dots + (n+1)$$

is an AP with n number of terms with common difference (-1) .

$$\begin{aligned} \therefore S_n &= \frac{n}{2} [a + l] \\ &= \frac{n}{2} [2n + (n+1)] \\ &= \frac{n(3n+1)}{2} \end{aligned}$$

3 Correct Solution.

2 Obtains $2^n \binom{2n}{n}$ by using

$$Z_0 = (1-i)^n.$$

1 Attempts to apply the result of part (i) (or) the sum provided.

END OF EXAM.